



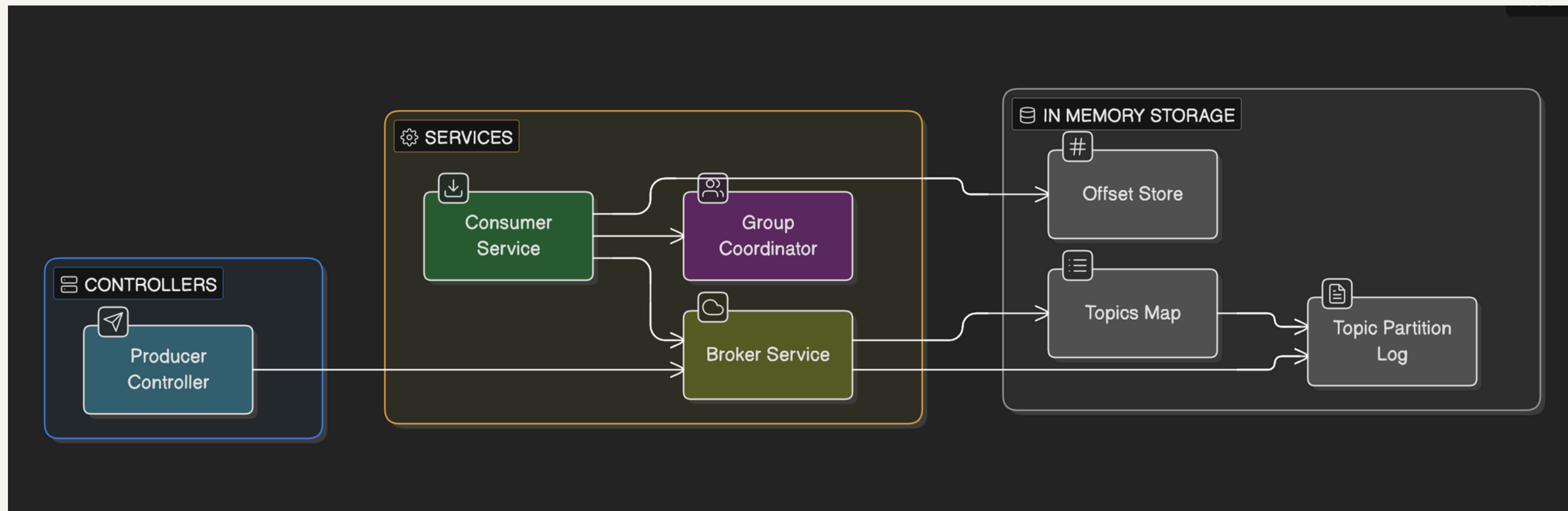
Day-5 BE

In-Memory Kafka Mock (Java + Spring Boot)



By Mani Rohit

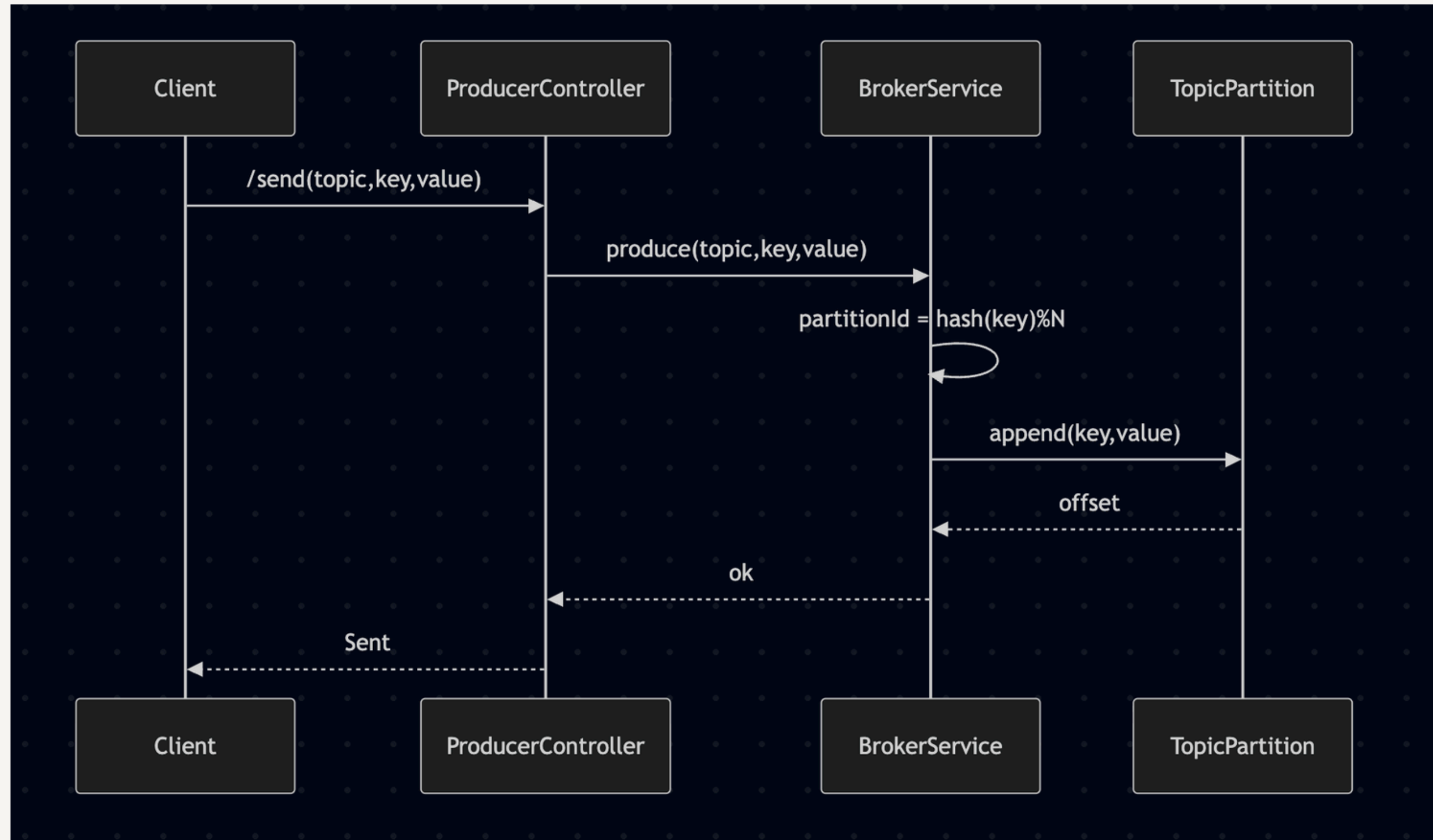
High-Level Architecture (HLD)



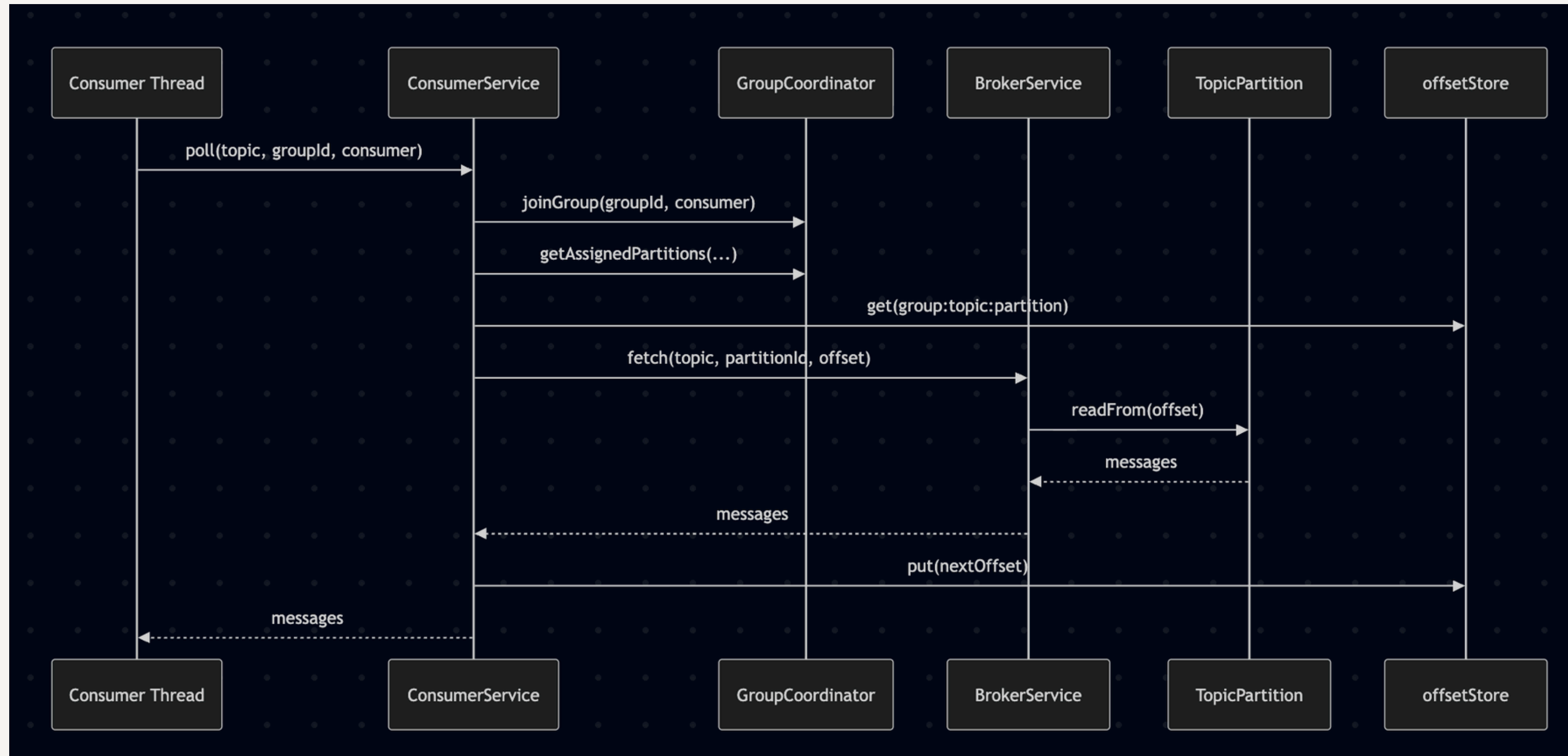
Data Model / Data Structures Used

- `Map<String, List<TopicPartition>> topics`
 - topic name → list of partitions
- `TopicPartition.log` = in-memory list of messages
 - preserves insertion order
- `offsetStore: Map<String, Integer>`
 - key: `"groupId:topic:partition"` → next offset to read
- `GroupCoordinator.groupMembers: Map<String, Set<String>>`
 - groupId → consumers (thread-safe set)

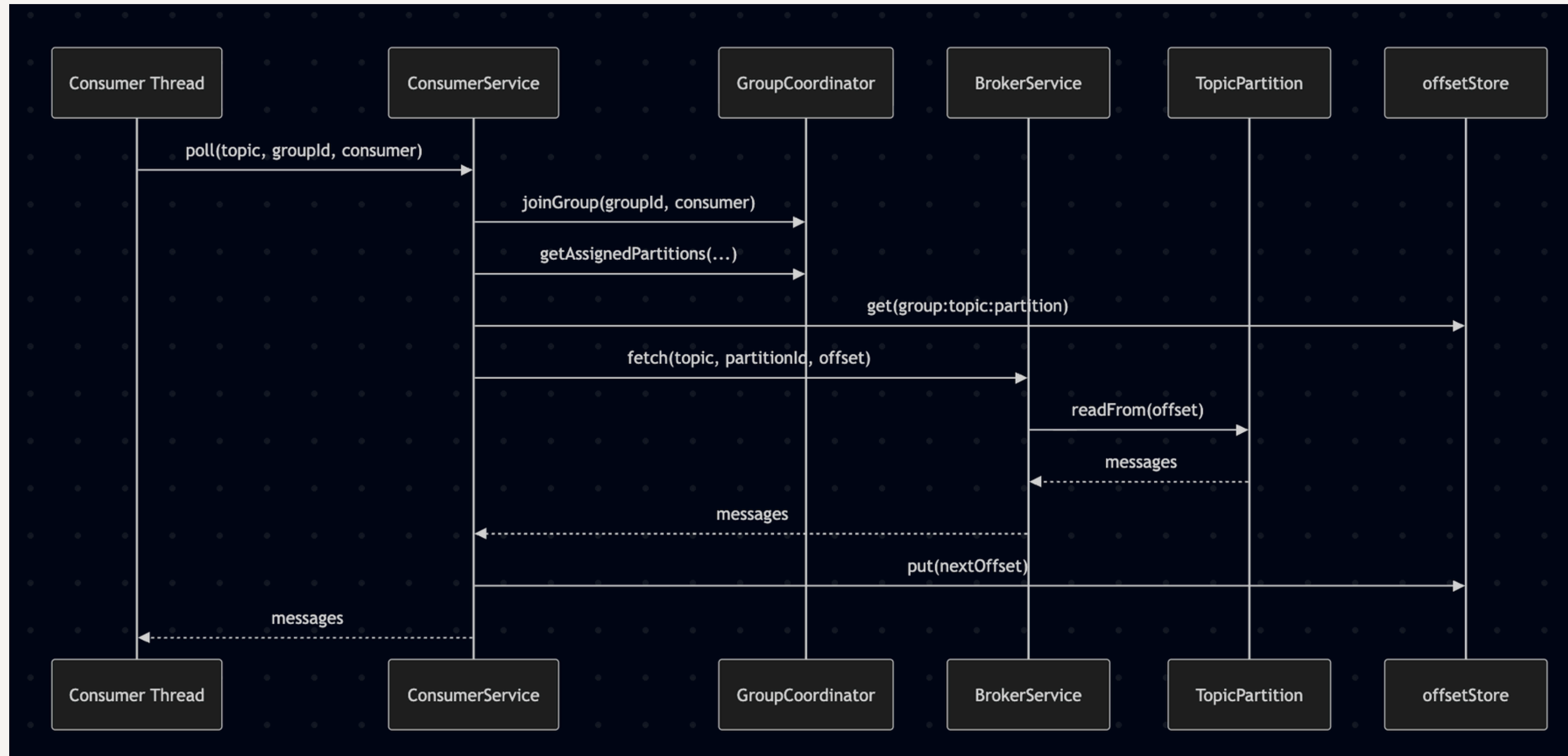
Producer Flow



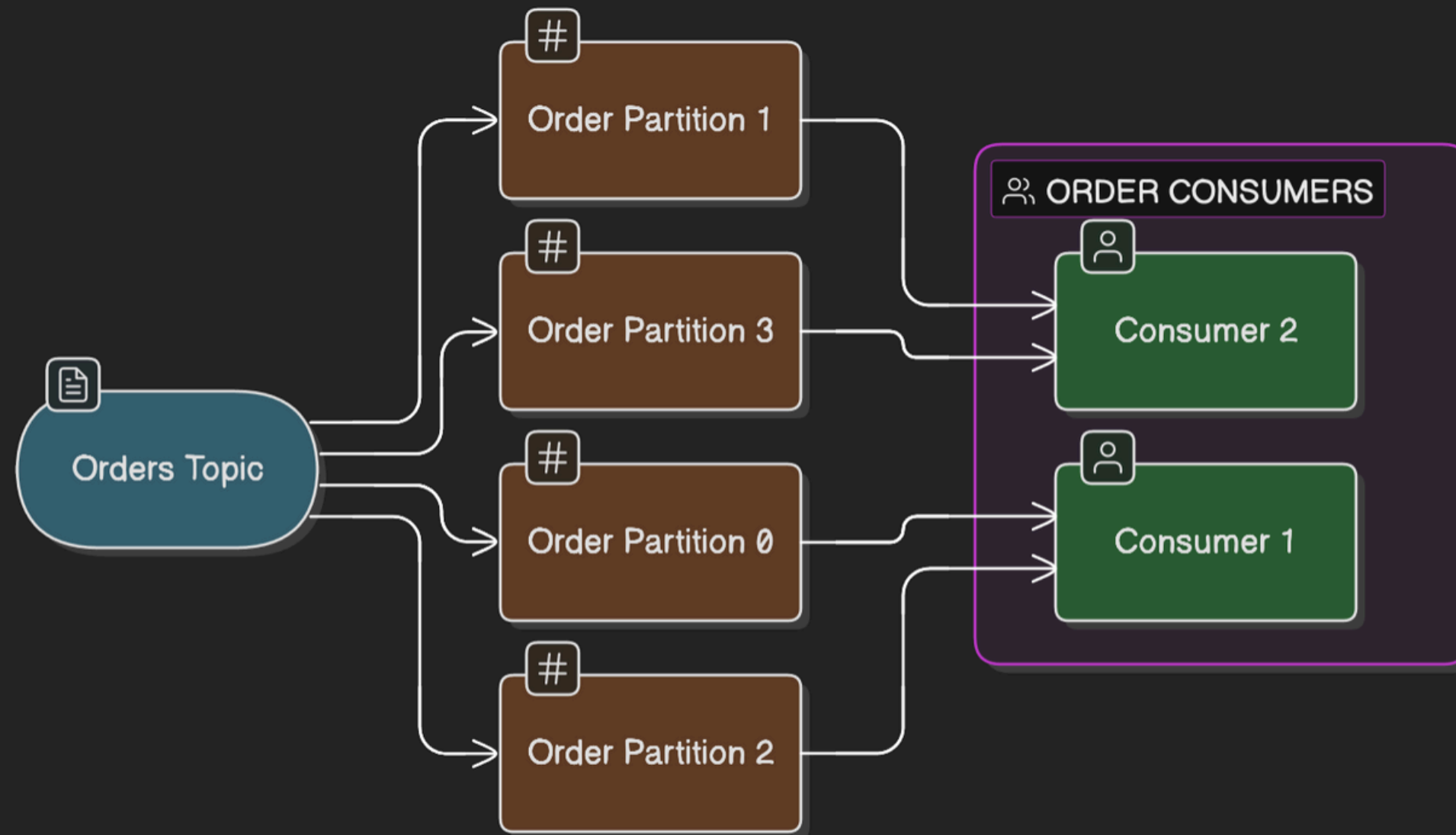
Consumer Flow (Poll + Offsets)



Consumer Flow (Poll + Offsets)



Consumer Groups & Partition Assignment



What I learned



- Why Kafka uses **append-only logs** and **stable offsets**
- Why ordering is only guaranteed **within a partition**
- How consumer groups enable **parallelism** while preventing duplicates per partition
- Why retention requires `baseOffset + nextOffset` (offsets must be monotonic)
- Threading model: consumer loops = poll → process → commit(offset) → repeat